Mikhail Razakov

Compiler / Static Analysis Engineer

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Compiler and static-analysis engineer working across LLVM optimization and program analysis, Roslyn data-flow analysis, compiler infrastructure, and backend/codegen projects.

Compiler engineering

MCST experience with LLVM optimization passes, loop analysis, interprocedural analysis, and legality for IR transforms.

Static analysis

SharpChecker at ISP RAS plus an independent Roslyn CFG fixed-point data-flow analyzer.

Backends & verification

SSA-like IR, liveness/interference, register allocation, x86-64 codegen, and differential execution.

EXPERIENCE

2026 — present

ISP RAS — SharpChecker

- Working on SharpChecker, ISP RAS's static-analysis platform for C#/.NET.

July — August 2026

MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass with loop analysis, side-effect/speculative-safety checks, and hoisting of loop-invariant instructions to preheaders.
- Developed conservative interprocedural global-IV analysis with APInt affine evolution, transitive effects, a separate legality phase, and LLVM IR transformation; unsupported CFG/call cases are rejected.

PROJECTS

LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

DerefAfterNullAnalyzer

Diagnostics for potentially unsafe dereference paths over the real CFG, with successor reprocessing until the data-flow state reaches a fixed point.

UniversalToolchain

The modular framework separates language composition from runtime execution and backs its contracts with executable verification/tests.

x86-64 Codegen & Register Allocation Lab

A reference interpreter and differential execution check generated-code semantics; the allocator assignment contract is verified independently.

SKILLS

Compilers

C++, LLVM, LLVM IR, optimization passes, interprocedural analysis, CFG, SSA, dominators, loop analysis, LICM, legality analysis

Static / Program Analysis

C#, .NET, Roslyn, ControlFlowGraph, fixed-point data-flow analysis, state propagation and joins

Codegen / Tooling

register allocation, liveness analysis, x86-64, CMake, Linux, GitHub Actions, differential testing

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

First Degree Diploma and Grand Prize “Perfection as Hope” for UniversalToolchain.

Moscow / remote